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Advanced 35 W Free-Piston Stirling Engine for Space Power Applications

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Abstract. This paper presents the projected performance and overall design characteristics of a high efficiency, low mass 35 W free-piston Stirling engine design. Overall (engine plus linear alternator) thermodynamic performance greater than 50% of Carnot, with a specific power close to 100 W/kg appears to be a reasonable goal at this small power level. Supporting test data and analysis results from exiting engines are presented. Design implications of high specific power in relatively low power engines is presented and discussed.

INTRODUCTION

Recently there is a renewed interest in free-piston Stirling engine (FPSE) converters for use in space power applications. The use of high efficiency FPSEs would allow a reduction in the radioisotope fuel by a factor of roughly four compared to existing RTGs. Free-piston Stirling engines and cryocoolers are demonstrating long life capability, and significant strides in performance of Stirling cycle machines have been made in the commercial sector. The following presents the characteristics and performance of existing engines, and presents an advanced design for a small engine designed for space power applications.

FPSE PERFORMANCE

The following plot presents the performance data available in the open literature for free-piston Stirling engines. This plot also includes test data from the current Sunpower EG-1000 engine. For comparison with the free-piston machines, performance of the Mod 2 automotive (kinematic) engine is also included. All machines use helium as the working fluid, except for the Mod 2, which uses hydrogen. This plot presents engine efficiency only (not including linear alternator efficiency) and is intended to show that engine efficiencies slightly exceeding 60 percent of Carnot are reasonably achievable.

The Curzon-Ahlborn efficiency curve (Curzon 1975) shown on the plot is the maximum power efficiency of an endoreversible heat engine system. This is not a limit to possible efficiency, but appears to be reasonable goal for real machines. The most efficient free-piston engine to date is seen to be the Sunpower EG-1000 engine. This is a prototype 1 kW machine designed for use in small domestic European cogeneration systems (Microgen 2001). Although this machine is designed for low cost commercial production, the engine achieves 58 percent of Carnot efficiency. With redesign it is expected that the machine would achieve greater than 60 percent of Carnot. Calculations for data points in the figure are as follows:

- MTI Mod 2 (Ernst 1997) efficiency was calculated by reducing P-V efficiency by measured friction losses. The peak efficiency upper bound occurs at 41% of maximum power point (lower bound).
- Sunpower EG-1000 efficiency was calculated by dividing P-V power by heat into the electrically-heated head. P-V power was calculated from measured electric power output using a known alternator efficiency of 85%.
- MTI CPTC efficiency is the average of heat-to-water and heat-to-head methods of calculating efficiency. Heat-to-head efficiency is 45% of Carnot efficiency and heat-to-water efficiency is about 54% of Carnot (Dhar 1997).
- MTI SPRE data was based on heat-to-water efficiency from the Dochat (1993) plot on page 92 of that report. The maximum efficiency (upper point) occurs at 35 % of design power (lower point)

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- Stirling Technology Company (STC) 55 W test data is from NASA (2001). Alternator efficiency was assumed to be 85%.
- Finally, the Sunpower 1979 model RE-1000 data point shown is from test point 602 in Schreiber (1986), which represents approximately the highest measured efficiency point.

Note that for both the Sunpower EG 1000 and the STC 55 W engine-only efficiency has been calculated assuming an 85% efficient alternator. The EG 1000 alternator efficiency is known. However if the efficiency of the STC alternator is different than the assumed 85% the engine efficiency will be different than the values presented on the plot.

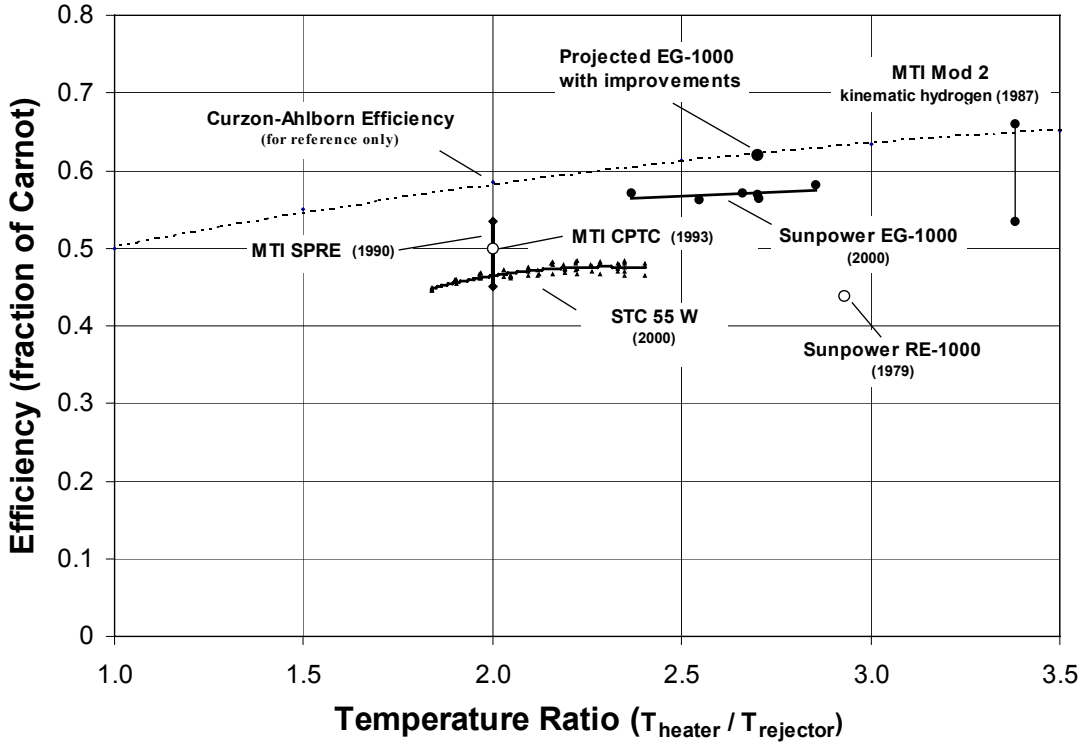


FIGURE 1. Comparison of Stirling Engine-Only Efficiencies (Curzon (1975); Dhar (1997); Dochat (1993); Ernst (1997), NASA (2002); Schreiber (1986)).

FPSE LIFE AND RELIABILITY DEMONSTRATIONS

Much of the ongoing life and reliability testing of free piston machines is proprietary. Some information on the long life capability of Stirling machines is publicly available and is summarized below.

A Sunpower M223 100 W Stirling-cycle refrigerator based on Sunpower's patented gas bearing technology has been on life test since 1995. This machine achieved 60,000 hours of maintenance free operation on September 9, 2002, and the life test is continuing. A similar M223 flew on the Space Shuttle (MacDonald 1994). STC has a 10 W Stirling engine, based on their flexure support technology, which achieved 66,000 degradation-free hours in 2002 (Qiu 2002).

SMALL ENGINE DESIGN

Under 2001 NASA SBIR Phase I funding, Sunpower optimized and designed the small FPSE converter shown in the following figure. In this design, two dynamically opposed engines would operate on the heat supplied by a single 250W General Purpose Heat Source (GPHS) module. The design is based on 120C reject temperature and uses an 83 percent efficient alternator to minimize mass. The design is projected to produce 33 W for 115W of heat input, and has a mass of only 305 gm.

Phase II of the SBIR that involves building and testing of the machine has just started. System optimization studies are underway in conjunction with NASA-Glenn, which at present are indicating that the system optimizes at a lower reject temperature and with a more efficient alternator. Reduction of the reject to 80C and the increase in alternator efficiency to 90.9 percent will result in an converter of approximately 400 gm producing approximately 40 W with the same 115 W heat input. The design presented here however is for the 33 W 305 gm machine. The final design will likely fall between 33 W to 40 W depending of the final system optimization. For now we are referring to the engine as an "Advanced 35 Watt Engine".

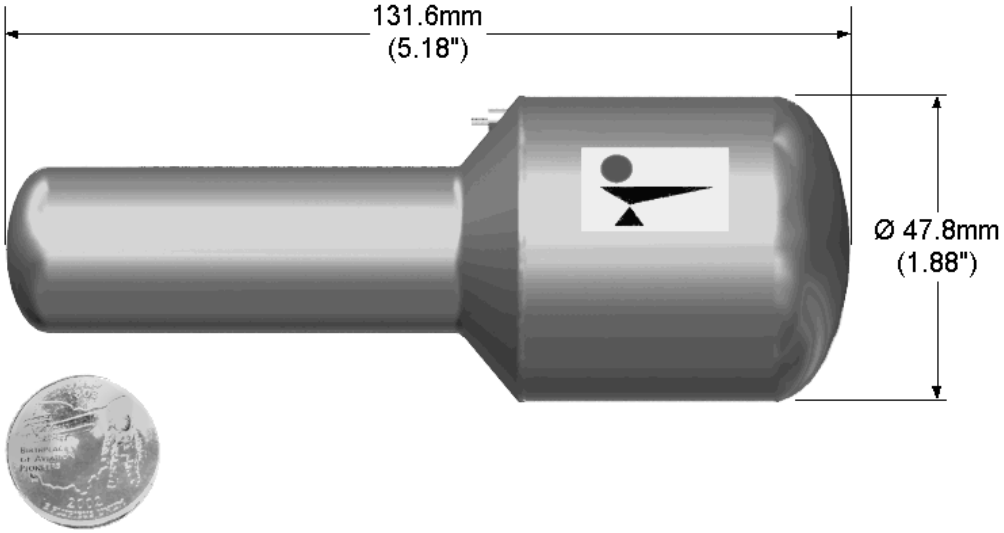


FIGURE 2. Sunpower Advanced 35 W Free-Piston Engine, Shown With Ohio State Quarter.

The following table presents some details of the machine designed under the Phase I effort. The engine (without alternator) is projected to have 60.8 percent of Carnot, which appears to be reasonable when considering the data presented in Figure 1. Design considerations that influence Stirling engine efficiency when scaling to small sizes will be discussed further below.

TABLE 1. Overall Performance and Design Parameters.

Parameter	Value
Electrical Power Output (W)	33.3
Hot End Metal Temperature (°C)	650
Rejector Metal Temperature (°C)	120
Projected Engine Efficiency (%)	34.9
Projected Engine % of Carnot Efficiency	60.8
Alternator Efficiency (%)	83
Projected Overall Efficiency (%)	29.0
Projected Overall % of Carnot Efficiency	50.5
Design Heat Input (W)	115
Total Mass (g)	305
Operating Temperature Ratio	2.35

Parameter	Value
Specific Power (W/kg)	109
Operating Frequency (hz)	100
Charge Pressure of helium (Mpa guage)	2.27
Piston Amplitude (mm)	4.0
Radial Running Clearances (microns)	6

The specific power projected for the design falls in the 100 W/kg range, which appears reasonable considering Sunpower's larger engine performance. The EG-1000 engine operates at 50 hertz and produces approximately 35 W/kg. Doubling the frequency of this machine would result in 70 W/kg. The existing engine also has flanges to allow tear-down and is also designed to meet stringent pressure vessel codes. Removal of the flanges and thinning the pressure vessel, as well as doubling the frequency, would put the EG-1000 in the 100 W/kg range. Thus specific powers around 100 W/kg appear reasonable.

EFFECTS ON EFFICIENCY WHEN SCALING DOWN IN POWER

When designing the engine presented here, Sunpower performed numerous computer optimizations to arrive at the best performance. Sunpower projects that the proposed engine will have higher efficiency than that of the EG1000, and there are good reasons to expect higher efficiency in a small engine. The primary reasons are outlined in the following

As engines are made smaller, the main difference is the increase in the ratio of the perimeter to cross sectional area of the machine. Perimeter varies directly with diameter, whereas cross sectional area varies with the square of the diameter. Power and associated heat flows into and out of the machine are proportional to the cross-sectional area of the machine. If all the other lengths of the machine are held constant, then the heat transfer per unit of surface area into and out of the machine decrease as the engine diameter is made smaller. Also as the diameter is decreased the wall thickness of the vessel can be made proportionally thinner for the same pressure to maintain the same stress level in the wall. Temperature drops through the wall therefore vary as the square of the diameter, with smaller machines having much reduced temperature drops. This is the primary reason that large machines resort to tubular or modular type heat exchangers, instead of simple monolithic heater heads where heat is transferred directly though the wall of the main vessel.

The EG-1000 has a monolithic heater head, as does the small engine presented here. However the heater wall temperature drop of the EG1000, due to its size, is 35 degrees C which in effect drops the temperature ratio of the internal gas. This wall temperature drop reduces the design point effective temperature ratio from 2.55 to 2.44. This results in an efficiency drop of more than 1 percentage point, or an efficiency penalty of more than 3 percent in relative terms. Overall nominal thermal efficiency of this machine is 30 percent.

For the small engine presented here, the temperature drop through the wall is very much less significant than in the high efficiency EG-1000 engine, which leads to higher efficiency. An additional factor that helps reduce this temperature drop results from the increase in frequency of the small machine. Increasing frequency reduces the required charge pressure of the machine, which in turn further reduces the required wall thickness of the vessel.

Because of the large total reductions in through the wall temperature drops, additional advantages are achieved because the lengths of the internal acceptor and rejector can be reduced. These components can then be optimized with length being very much independent to wall temperature drop effects. Typically the reduced lengths reduce the internal heat exchanger surface areas, which in turn reduce thermal hysteresis losses arising from the pressure swing of the machine.

The most effective and useful part of the internal acceptor and rejector of an engine are those areas adjacent to the expansion space and compression space, respectively. These flow entry regions see the largest wall-to-gas temperature difference and additionally have the highest rates of heat transfer because the flow is developing in these regions. Typically these heat exchangers want to optimize at rather short lengths if wall temperature drops are not significant. Small engines thus allow much freer optimization of the heat exchangers.

One might suspect that seal losses would become significant in small sizes, because of the increase in the ratio of perimeter to area of the machine. However this turns out not to be a serious problem. Assuming that the piston stroke and seal length remain fixed, it is instructional to look at the ratio of seal power loss to engine power. Seal power loss varies as the diameter times the cube of the clearance gap, while engine power scales with the square of the piston diameter. To maintain the seal loss at a fixed percentage of the piston power thus requires that the gap vary as the diameter to the 1/3 power. Again advantage can be made in this area by increasing frequency while reducing pressure to maintain engine power. Leakage power varies as the square of the pressure, thus reducing the pressure has a significant effect on the leakage.

EFFICIENCY, MASS, AND REJECTION TEMPERATURE TRADEOFFS

Rejection temperature has a significant effect on the FPSE converter efficiency and power. Alternator efficiency is largely a function of its mass. The design presented here has a 120C reject and utilizes an 83 percent efficient alternator.

The following graph shows the relationship of output power at different machine masses (primarily driven by the alternator mass) as a function of reject temperatures. Note that in this plot that the heat input is fixed so that electrical power as presented is directly proportional to the conversion efficiency.

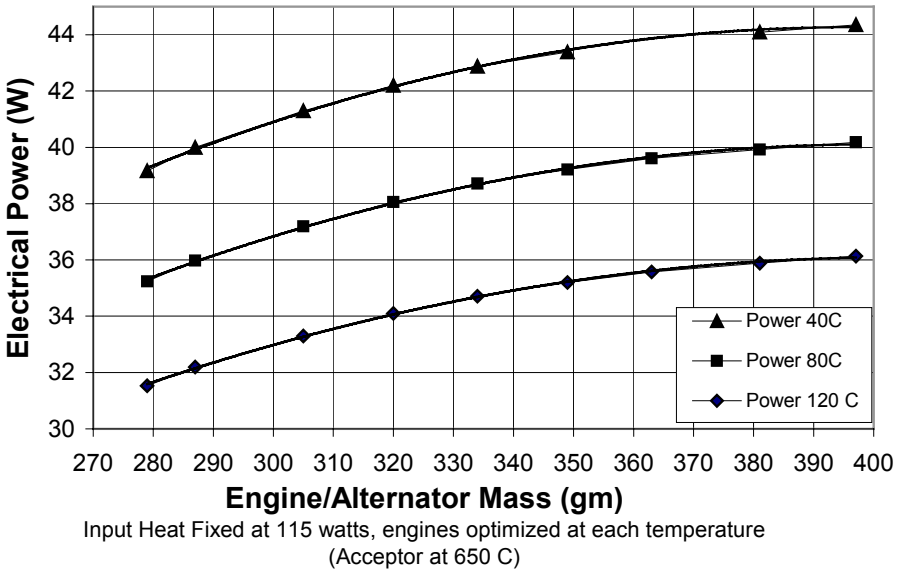


FIGURE 3. Electrical Power Versus Converter Mass at Different Rejector Temperatures.

Data presented in the above plot is currently being input by NASA-Glenn into a total system model to determine the optimal rejection temperature as well as converter mass. Early indications from that effort are that the system optimizes near the 80C reject point, and at a mass in the vicinity of 400 grams. As seen in the above plot, the engine alternator combination at that point will produce 40 W for a mass of 400 gm thus resulting in a specific power of 100 W/kg.

SUMMARY

This paper presents the design of an advanced small 35 Watt free-piston Stirling engine with integral linear alternator for space power applications. Notable features of the design are high thermal to electric energy conversion efficiencies (exceeding 50 percent of Carnot) and high specific power (~100 W/kg). The predictions in efficiency and power are reasonable considering recently achieved gains in performance of the larger Sunpower EG-1000 engine. As discussed in the paper, there are significant efficiency advantages when scaling Stirling engines

down in power. Small Stirling engines, as presented here, should have very high thermodynamic performance while maintaining the simplicity and structural reliability of monolithic heater head construction.

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